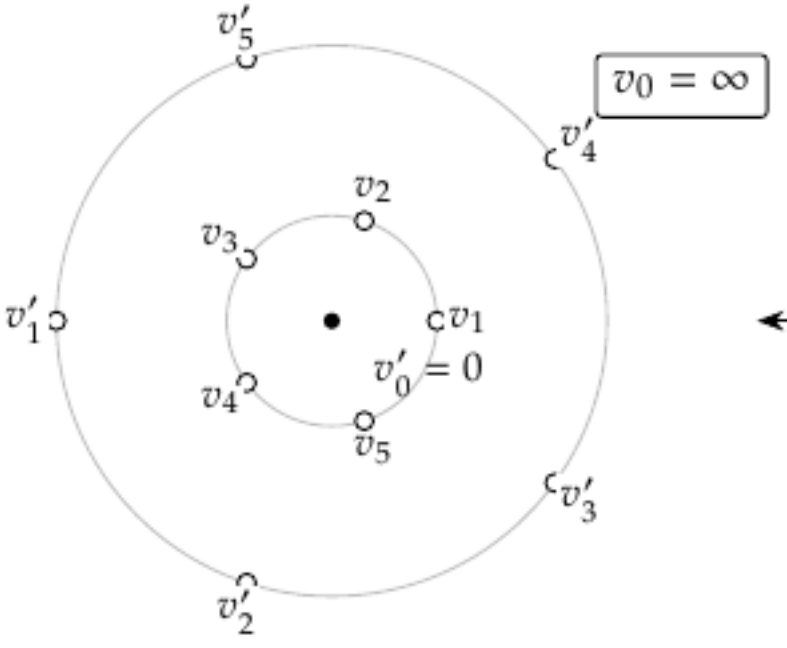
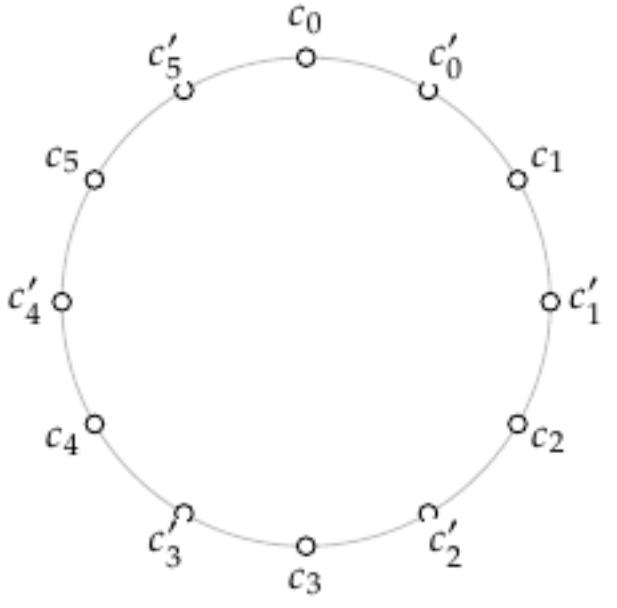
The Klein divisor in $\mathbb{P}^1$



chosen coordinate

The twelve cusps of $X(5)$



$$v_i = \varphi^{-1}\zeta^{i-1}, \quad v'_i = -\varphi\zeta^{i-1}$$
$$1 \le i \le 5, \quad \zeta = e^{2\pi i/5}$$

$$c_i \mapsto v_i, \quad c'_i \mapsto v'_i$$

Schematic arrangement of the cusps.

$$M_2(\Gamma(5)) \xrightarrow{\text{res}} \{(r_p)_{p\in D} : \textstyle\sum r_p = 0\}, \qquad \dim = 11.$$